

Curriculum Vitae

Main | CV | Research | Teaching | Awards

Education

Ph.D. California Institute of Technology (USA), Computer Science

Dissertation: *SPICE2 - A Spatial Parallel Architecture for Accelerating the SPICE Circuit Simulator*

Degree Conferred: September 2010 [Link]

M.S., California Institute of Technology (USA), Computer Science

Degree Conferred: June 2006

Thesis: *Packet-Switched FPGA-Overlay Networks* [Link]

M.S., California Institute of Technology (USA), Electrical Engineering

Degree Conferred: June 2005

B.E., University of Pune (India), Electronics and Telecommunication Engineering

Project: *FPGA-based testing system for Siemens Railway Signalling Relays* [Link]

[slides] [guide]

Degree Conferred: August 2002

Research Interests

Concurrent and Spatial Architectures, Parallel Processing, Heterogeneous Architectures and Compilation Tools, Communication-Centric Design

Selected Grants

NSERC Discovery Grant Renewal (2024–2029), *Rethinking Configurable Logic Architectures in the Age of AI*, CAD \$265K

Mitacs Globalink (2020–2021), *High-Level Synthesis of Nengo Neural Networks*, CAD \$15K

Huawei Research Grant (PI subcontract; University of Toronto lead institution) (2019–2021), *CGRA and Embedded FPGA Architecture and Tools*, CAD \$157.5K

Research Tools and Instruments Grant (Co-PI) (2019–2020), *Real-Time Baseband Signal Processing and Reconfigurable Computing Development Platform*, CAD \$148,905

Xilinx Unrestricted Research Gift (2018–2019), CAD \$10K

NSERC Discovery Grant (2017–2022), *Enhancing Hardware Compilation for Reconfigurable Architectures using Machine Learning and Cloud Computing*, CAD \$165K

University of Waterloo Startup Grant (2016–present), CAD \$150K

AcRF Tier 1 Grant (Nov 2015) S\$100K (1 year)

Delta Electronics Grant (Co-PI) (August 2015) S\$100K (2 years)

MIT SMART Innovation Grant (Co-PI) (August 2015) S\$60K (2 years)

NTU CELT Excellence in Education Grant (November 2014) S\$37K (1 year)

AcRF Tier 1 Grant (March 2014) S\$150K (2 years)

NTU CELT Excellence in Education Grant (October 2013) S\$40K (1 year)

NTU CELT Excellence in Education Grant (March 2013) S\$25K (1 year)

NTU CoE Competitive Seed Grant S\$50K (Jan-March 2013)

NTU SCE Startup Grant S\$100K (3 years)

Preprints

[PDF]

“A Protocol-Independent Transport Architecture”

Kimiya Mohammadtaheri, David Gao, Samuel Zhang, Matthew Chen, Eric Su, Pengyu Ji, Saad Syed, Chris

Journal Publications

[PDF]

“HopliteML: Evolving application customized FPGA NoCs with adaptable routers and regulators”

Gurshaant Malik, Ian Elmor Lang, Rodolfo Pellizzoni, Nachiket Kapre

ACM Transactions on Reconfigurable Technology and Applications (Special Issue FPL 2020)

[PDF]

“RapidLayout: Fast Hard Block Placement of FPGA-optimized Systolic Arrays using Evolutionary Algorithms” (ACM TRETS 2023 Best Paper Award)

Niansong Zhang, Xiang Chen, Nachiket Kapre

ACM Transactions on Reconfigurable Technology and Applications (Special Issue FPL 2020)

[PDF]

“HopliteBuf: Network Calculus-Based Design of FPGA NoCs with Provably Stall-Free FIFOs”

Tushar Garg, Saud Wasly, Rodolfo Pellizzoni, Nachiket Kapre

ACM Transactions on Reconfigurable Technology and Applications (Special Issue FPGA 2019)

[PDF]

CaffePresso: Accelerating Convolutional Networks on Embedded SoCs

Gopalakrishna Hegde, Siddhartha, Nachiket Kapre

ACM Transactions on Embedded Computing Systems (Special Issue on ESWEEK 2016)

[PDF]

“Hoplite: A Deflection-Routed Directional Torus NoC for FPGAs” (ACM TRETS 2017 Best Paper Award)

Nachiket Kapre, Jan Gray

ACM Transactions on Reconfigurable Technology and Applications (Special Issue FPL 2015)

[PDF]

“Optimizing Soft Vector Processing in FPGA-based Embedded Systems”

Nachiket Kapre

ACM Transactions on Reconfigurable Technology and Applications (Special Issue FPL 2014), Published: 2016

[PDF]

“A Case for Embedded FPGA-based SoCs in Energy-Efficient Acceleration of Graph Problems”

Pradeep Moorthy, Nachiket Kapre

Supercomputing Frontiers and Innovations (Special Best Papers Issue from 2015 Supercomputing Frontiers Conference), Published: 2016

[PDF]

“Communication Optimization of Iterative Sparse Matrix-Vector Multiply on GPUs and FPGAs”

Abid Rafique, George Constantinides, Nachiket Kapre

IEEE Transactions on Parallel and Distributed Systems, Jan 2015

[PDF]

“SPICE2: Spatial Processors Interconnected for Concurrent Execution for accelerating the SPICE Circuit Simulator using an FPGA”

Nachiket Kapre and André DeHon

Transactions in CAD (Special Issue on Parallel CAD), Volume 31 Issue 1 January 2012

[PDF]

“Spatial Hardware Implementation for Sparse Graph Algorithms in GraphStep”

Michael deLorimier, Nachiket Kapre, Nikil Mehta and André DeHon

ACM Transactions on Autonomous and Adaptive Systems: Spatial Computing
Special Issue, September 2011

[PDF]

“An NoC Traffic Compiler for efficient FPGA implementation of Sparse Graph-Oriented Workloads”

Nachiket Kapre and André DeHon

International Journal of Reconfigurable Computing Volume 2011 Article ID 745147

[PDF]

“Pipelined Saturated Accumulation”

Karl Papadantonakis, Nachiket Kapre, Stephanie Chan, and André DeHon

IEEE Transactions on Computers, February 2009.

Conference/Workshop Publications (Full papers)

[PDF]

“Jack of All Scales: A Versatile FPGA Tensor Block for MXFP Precisions”

Marwan Mekhemer, Ahmed Elsousy, Balaji Venkatesh, Raphael Rowley, Vaughn Betz, Nachiket Kapre, Andrew Boutros

International Conference on Field-Programmable Logic and Applications, Sep 2026

[PDF]

“Cocotb-PYNQ-PR: From Co-Simulation to Deployment, A Unified DFX Framework for PYNQ”

Emir Guevara, Chaitanya Sharma, Gavin Lusby, Nachiket Kapre

International Conference on Field-Programmable Logic and Applications, Sep 2026

(Stamatis Vassiliadis Memorial Best Paper Award)

[PDF]

“GraphNoC: Graph Neural Networks for Application-Specific FPGA NoC Performance Prediction”

Gurshaant Malik, Nachiket Kapre

International Conference on Field-Programmable Technology, Dec 2024

(Best Paper Award)

[PDF]

“Ditty: Directory-based Cache Coherence for Multicore Safety-critical Systems”

Zhuanhao Wu, Marat Bekmyrza, Nachiket Kapre, Hiren Patel

Design, Automation, and Test in Europe, Apr 2023

(Best Paper Nominee)

[PDF]

“Managing HBM Bandwidth on Multi-Die FPGAs with FPGA Overlay NoCs”

Srinirdheeshwar Kuttuva Prakash, Hiren Patel, Nachiket Kapre

International Conference on Field-Programmable Custom Computing Machines, May 2022

[PDF]

“Mocarabe: High-Performance Time-Multiplexed Overlays for FPGAs”

Frederick Tombs, Alireza Mellat, Nachiket Kapre

International Conference on Field-Programmable Custom Computing Machines, May 2021

[PDF]

“RapidLayout: Fast Hard Block Placement of FPGA-optimized Systolic Arrays using Evolutionary Algorithms”

Niansong Zhang, Xiang Chen, Nachiket Kapre

International Conference on Field-Programmable Logic and Applications, Sep 2020 (Nominee for Michal Servit Memorial Award)

[PDF]

“Learn the Switches: Evolving FPGA NoCs with Stall-Free and Backpressure based Routers”
Gurshaant Malik, Ian Lang, Rodolfo Pellizzoni, Nachiket Kapre
International Conference on Field-Programmable Logic and Applications, Sep 2020 (Nominee for Stamatis Vassiliadis Memorial Award)

[PDF]

“Partitioning Systolic Arrays for Fun and Profit” (Best Paper Award)

Harry Chan, Gurshaant Malik, Nachiket Kapre
International Conference on Field-Programmable Technology, Dec 2019

[PDF]

“Scaling the Cascades: Interconnect-aware FPGA implementation of Machine Learning problems”

Ananda Samajdar, Tushar Garg, Tushar Krishna, Nachiket Kapre
29th International Conference on Field-Programmable Logic and Applications, Sep 2019

[PDF]

“Timing-aware routing in the RapidWright framework”

Leo Liu, Nachiket Kapre
29th International Conference on Field-Programmable Logic and Applications, Sep 2019

[PDF]

“Enhancing Butterfly Fat Tree NoCs for FPGAs with lightweight flow control”

Gurshaant Malik, Nachiket Kapre
International Conference on Field-Programmable Custom Computing Machines, Apr 2019

[PDF]

“HopliteBuf: FPGA NoCs with Provably Stall-Free FIFOs” (Best Paper Nominee)

Tushar Garg, Saud Al Wasly, Rodolfo Pellizzoni, Nachiket Kapre
International Symposium on Field-Programmable Gate Arrays, Feb 2019

[PDF]

“Implementing NEF Neural Networks on Embedded FPGAs”

Benjamin Morcos, Terrence Stewart, Chris Eliasmith, Nachiket Kapre
International Conference on Field-Programmable Technology, December 2018

[PDF]

“DaCO: A High-Performance Token Dataflow Coprocessor Overlay for FPGAs”

Siddhartha, Nachiket Kapre
International Conference on Field-Programmable Technology, December 2018

[PDF]

“FastTrack: Leveraging Heterogeneous FPGA Wires to Design Low-cost High-performance Soft NoCs”

Nachiket Kapre, Tushar Krishna
International Symposium on Computer Architecture, June 2018

[PDF]

“LegUp-NoC: High-Level Synthesis of Loops with Indirect Addressing”

Asif Islam, Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, April-May 2018

[PDF]

“HopliteQ: Priority-Aware Routing in FPGA Overlay NoCs”

Siddhartha, Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, April-May 2018

- [PDF]
“HopliteRT: An Efficient FPGA NoC for Real-Time Applications”
 Saud Al Wasly, Rodolfo Pellizzoni, Nachiket Kapre
International Conference on Field-Programmable Technology, December 2017
- [PDF]
“Deflection Routed Butterfly Fat Trees on FPGAs”
 Nachiket Kapre
27th International Conference on Field-Programmable Logic and Applications, Sep 2017
- [PDF]
“Enabling Partial Reconfiguration and Low Latency Routing using Segmented FPGA NoCs”
 Kizhepatt Vipin, Jan Gray, Nachiket Kapre
27th International Conference on Field-Programmable Logic and Applications, Sep 2017
- [PDF]
“On Bit-Serial NoCs for FPGAs”
 Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, May 2017
- [PDF]
“Implementing FPGA overlay NoCs using the Xilinx UltraScale memory cascades”
 Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, May 2017
- [PDF]
“eBSP: Managing NoC traffic for BSP workloads on the 16-core Adapteva Epiphany-III Processor”
 Siddhartha, Nachiket Kapre
Design, Automation, and Test in Europe, March 2017
- [PDF]
“Deflection Routing for Multi-Level FPGA Overlay NoCs”
 Chethan Kumar H B, Shubham Agarwal, Nachiket Kapre
International Conference on Field-Programmable Technology, December 2016
- [PDF]
“Preventive Detection of Mosquito Populations using Embedded Machine Learning on Low Power IoT Platforms”
 Prashant Ravi, Uma Syam, Nachiket Kapre
Seventh ACM Symposium on Computing and Development, Nov 2016
- [PDF]
“CaffePresso: An Optimized Library for Deep Learning on Embedded Accelerator-based platforms” (Best Paper Award)
 Gopalakrishna Hegde, Siddhartha, Nachiappan Ramasamy, Nachiket Kapre
International Conference on Compilers, Architecture, and Synthesis for Embedded Systems, Oct 2016
- [PDF]
“Hoplite-DSP: Harnessing the Xilinx DSP48 Multiplexers to efficiently support NoCs on FPGAs”
 Chethan Kumar H B, Nachiket Kapre
26th International Conference on Field-Programmable Logic and Applications, Sep 2016
- [PDF]
“Boosting Convergence of Timing Closure using Feature Selection in a Learning-Driven Approach”
 Que Yanghua, Harnhua Ng, Nachiket Kapre
26th International Conference on Field-Programmable Logic and Applications, Sep 2016

- [PDF]
“Survey of Domain-Specific Languages for FPGA Computing”
 Nachiket Kapre, Samuel Bayliss
26th International Conference on Field-Programmable Logic and Applications, Sep 2016
- [PDF]
“Marathon: Statically-Scheduled Conflict-Free Routing on FPGA Overlay NoCs”
 Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, May 2016
- [PDF]
“GPU-Accelerated High-Level Synthesis for Bitwidth Optimization of FPGA Datapaths”
 Nachiket Kapre, Ye Deheng
International Symposium on Field-Programmable Gate Arrays, Feb 2016
- [PDF]
“Hoplite: Building Austere Overlay NoCs for FPGAs” (Best Paper Award)
 Nachiket Kapre, Jan Gray
25th International Conference on Field-Programmable Logic and Applications, Sep 2015
- [PDF]
“Limits of FPGA Acceleration of 3D Green’s Function Computation for Geophysical Applications”
 Nachiket Kapre, Selvakumar Jayakrishnan, Parjanya Gupta, Sagar Masuti, Sylvain Barbot
25th International Conference on Field-Programmable Logic and Applications, Sep 2015
- [PDF]
“Custom FPGA-based Soft-Processors for Sparse Graph Acceleration”
 Nachiket Kapre
26th IEEE International Conference on Application-specific Systems, Architectures and Processors, July 2015
- [PDF]
“GraphMMU: Memory Management Unit for Sparse Graph Accelerators”
 Nachiket Kapre, Han Jianglei, Andrew Bean, Pradeep Moorthy, Siddhartha
22nd Reconfigurable Architectures Workshop, 2015 (co-located with IPDPS 2015), May 2015
- [PDF]
“Enhancing Speedups for FPGA Accelerated SPICE through Frequency Scaling and Precision Reduction”
 Lim Hui Hui, Nachiket Kapre
22nd Reconfigurable Architectures Workshop, 2015 (co-located with IPDPS 2015), May 2015
- [PDF]
“Zedwulf: Power-Performance Tradeoffs of a 32-node Zynq SoC cluster”
 Pradeep Moorthy, Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, May 2015
- [PDF]
“Driving Timing Convergence of FPGA Designs through Machine Learning and Cloud Computing”
 Nachiket Kapre, Bibin Chandrashekar, Harnhua Ng, Kirvy Teo
International Symposium on Field-Programmable Custom Computing Machines, May 2015
- [PDF]
“Energy-Efficient Acceleration of OpenCV Saliency Computation using Soft Vector Processors”
 Gopalakrishna Hegde, Nachiket Kapre
International Symposium on Field-Programmable Custom Computing Machines, May 2015

[PDF]

“On Data Forwarding in Deeply Pipelined Soft Processor”

Hui Yan Cheah, Suhaib A. Fahmy and Nachiket Kapre

International Symposium on Field-Programmable Gate Arrays, February 2015

[PDF]

“Relax-Miracle: GPU Parallelization of Semi-Analytic Fourier-Domain solvers for Earthquake Modeling”

Sagar Masuti, Sylvain Barbot, and Nachiket Kapre

International Conference on High Performance Computing, December 2014

[PDF]

“Comparing Soft and Hard Vector Processing in FPGA-based Embedded Systems” (Best Paper Nominee)

Soh Jun Jie, and Nachiket Kapre

International Conference on Field-Programmable Logic and Applications, September 2014

[PDF]

“Limits of Statically Scheduled Token Dataflow Processing”

Nachiket Kapre, and Siddhartha

4th International Workshop on Data-Flow Execution Models for Extreme Scale Computing (co-located with PACT 2014), August 2014

[PDF]

“System-Level FPGA Device Driver with High-Level Synthesis Support”

Vipin Kizhepatt, Shreejit Shanker, Dulitha Gunasekara, Suhaib A Fahmy, Nachiket Kapre

International Conference on Field-Programmable Technology, December 2013

[PDF]

“Exploiting Input Parameter Uncertainty for Reducing Datapath Precision of SPICE Device Models”

Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, April 2013

[PDF]

“Application Composition and Communication Optimization of Iterative Solvers using FPGAs” (HiPEAC Paper Award)

Abid Rafique, Nachiket Kapre and George Constantinides

International Symposium on Field-Programmable Custom Computing Machines, April 2013

[PDF]

“Enhancing Performance of Tall-Skinny QR factorization using FPGAs”

Abid Rafique, Nachiket Kapre and George Constantinides

International Conference on Field-Programmable Logic and Applications, August 2012

[PDF]

“FX-SCORE: A Framework for Fixed-Point Compilation of SPICE Device Models using Gappa++”

Helene Martorell and Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, April 2012

[PDF]

“A High Throughput FPGA-based Implementation of the Lanczos Method for the Symmetric Extremal Eigenvalue Problem”

Abid Rafique, Nachiket Kapre, and George Constantinides

International Symposium on Applied Reconfigurable Computing, March 2012

[PDF]

“VLIW-SCORE: Beyond C for Sequential Control of SPICE FPGA Acceleration” (Best Paper

Award)

Nachiket Kapre and André DeHon

International Conference on Field-Programmable Technology, December 2011

[PDF]

“SPICE2 - A Spatial Parallel Architecture for Accelerating the SPICE Circuit Simulator”

Nachiket Kapre and André DeHon

The First Workshop on the Intersections of Computer Architecture and Reconfigurable Logic, December 2010

[PDF]

“An NoC Traffic Compiler for efficient FPGA implementation of Sparse Graph-Oriented Workloads”

Nachiket Kapre and André DeHon

Reconfigurable Communication-centric Systems on Chip, May 2010

[PDF]

“Parallelizing Sparse Matrix-Solve for SPICE Circuit Simulation using FPGAs”

Nachiket Kapre and André DeHon

International Conference on Field-Programmable Technology, December 2009

[PDF]

“Performance Comparison of Single-Precision SPICE Model-Evaluation on FPGA, GPU, Cell, and Multi-Core Processors”

Nachiket Kapre and André DeHon

International Conference on Field-Programmable Logic and Applications, September 2009

[PDF]

“Accelerating SPICE Model-Evaluation using FPGAs”

Nachiket Kapre and André DeHon

IEEE Symposium on Field-Programmable Custom Computing Machines, April 2009

[PDF]

“Optimistic Parallelization of Floating-Point Accumulation”

Nachiket Kapre and André DeHon

IEEE Symposium on Computer Arithmetic, June 2007.

[PDF]

“Packet-Switched vs. Time-Multiplexed FPGA Overlay Networks” (FCCM20 25-most influential papers award winner)

Nachiket Kapre, Nikil Mehta, Michael deLorimier, Raphael Rubin, Henry Barnor, Michael Wilson, Michael Wrighton and André DeHon

IEEE Symposium on Field-Programmable Custom Computing Machines, April 2006.

[PDF]

“GraphStep: A System Architecture for Sparse Graph Algorithms”

Michael deLorimier, Nachiket Kapre, Nikil Mehta, Dominic Rizzo, Ian Eslick, Raphael Rubin, Tomas Uribe, Thomas Knight Jr., and André DeHon

IEEE Symposium on Field-Programmable Custom Computing Machines, April 2006.

[PDF]

“Pipelined Saturated Accumulation”

Karl Papadantonakis, Nachiket Kapre, Stephanie Chan, and André DeHon

International Conference on Field-Programmable Technology, December 2005.

[PDF]

“Design Patterns for Reconfigurable Computing”

André DeHon, Joshua Adams, Michael deLorimier, Nachiket Kapre, Yuki Matsuda, Helia Naeimi, Michael Vanier, and Michael Wrighton

IEEE Symposium on Field-Programmable Custom Computing Machines, April 2004.

Conference Publications (Short Papers)

[PDF]

“Cocotb-Pynq: Co-simulating Python+RTL applications targeting Pynq platforms with Cocotb”

Gavin Lusby, Nachiket Kapre

International Conference on Field-Programmable Logic and Applications, Sep 2025

[PDF]

“Worst-Case Latency Analysis for the Versal NoC Network Packet Switch”

Ian Lang, Nachiket Kapre, Rodolfo Pellizzoni

IEEE/ACM Symposium on Networks-on-Chip, Oct 2021

[PDF]

“Exploring the Impact of Switch Arity on Butterfly Fat Tree FPGA NoCs”

Ian Lang, Ziqiang Huang, Nachiket Kapre

International Conference on Field-Programmable Custom Computing Machines, Apr 2020

[PDF]

“RapidRoute: Fast Assembly of Communication Structures for FPGA Overlays” (Best Short Paper Runner-up)

Leo Liu, Jay Weng, Nachiket Kapre

International Conference on Field-Programmable Custom Computing Machines, Apr 2019

[PDF]

“Applying Models of Computation to OpenCL Pipes for FPGA Computing”

Nachiket Kapre, Hiren Patel

5th International Workshop on OpenCL, May 2017

[PDF]

“120-core microAptiv MIPS Overlay for the Terasic DE5-NET FPGA board”

Chethan Kumar H B, Gourav Modi, Prashant Ravi, Nachiket Kapre

International Symposium on Field-Programmable Gate Arrays, Feb 2017

[PDF]

“Vector Acceleration of 1-D DWT Computations using Sparse Matrix Skeletons”

Sidharth Maheshwari, Gourav Modi, Siddhartha, Nachiket Kapre

26th International Conference on Field-Programmable Logic and Applications, Sep 2016

[PDF]

“Improving Classification Accuracy of a Machine Learning approach for FPGA Timing Closure”

Que Yanghua, Nachiket Kapre, Harnhua Ng, Kirvy Teo

International Symposium on Field-Programmable Custom Computing Machines, May 2016

[PDF]

“Case for Design-Specific Machine Learning in Timing Closure of FPGA Designs”

Que Yanghua, Chinnakkannu Adaikkal Raj, Harnhua Ng, Kirvy Teo, and Nachiket Kapre

International Symposium on Field-Programmable Gate Arrays, Feb 2016

[PDF]

“InTime: A Machine Learning Approach for Efficient Selection of FPGA CAD Tool Parameters”

Nachiket Kapre, Harnhua Ng, Kirvy Teo and Jaco Naude

International Symposium on Field-Programmable Gate Arrays, February 2015

[PDF]

“Fanout Decomposition Dataflow Optimizations for FPGA-based Sparse LU Factorization”

Siddhartha, and Nachiket Kapre

International Conference on Field-Programmable Technology, December 2014

[PDF]

“Analysis and Optimization of a Deeply Pipelined FPGA Soft Processor”

Hui Yan Cheah, Suhaib A. Fahmy and Nachiket Kapre

International Conference on Field-Programmable Technology, December 2014

[PDF]

“Heterogeneous Dataflow Architectures for FPGA-based Sparse LU Factorization”

Siddhartha, and Nachiket Kapre

International Conference on Field-Programmable Logic and Applications, September 2014

[PDF]

“Breaking Sequential Dependencies in FPGA-based Sparse LU Factorization”

Siddhartha, and Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2014

[PDF]

“MixFX-SCORE: Heterogeneous Fixed-Point Compilation of Dataflow Computations”

Ye Deheng, and Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2014

[PDF]

“Timing Fault Detection in FPGA-based Circuits”

Edward Stott, Joshua M. Levine, Peter Y. K. Cheung, and Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2014

Posters

“Evaluating Embedded FPGA Accelerators for Deep Learning Applications”

Gopalakrishna Hegde, Siddhartha, Nachiappan Ramasamy, Vamsi Buddha, Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2016

“Communication Optimization for the 16-core Epiphany Floating-Point Processor Array”

Siddhartha, Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2016

“Machine-Learning driven Auto-Tuning of High-Level Synthesis for FPGAs”

Li Ting, Harri Supto Wijaya, and Nachiket Kapre

International Symposium on Field-Programmable Gate Arrays, Feb 2016

“Sparse Graph Processing using Soft-Processors”

Nachiket Kapre

International Symposium on Field-Programmable Custom Computing Machines, May 2015

“FPGA Acceleration of Irregular Iterative Computations using Criticality-Aware Dataflow Optimizations”

Siddhartha, and Nachiket Kapre

International Symposium on Field-Programmable Gate Arrays, February 2015

[PDF]

“Measuring Timing Errors in FPGA-based Circuits”

Joshua Levine, Edward Stott, and Nachiket Kapre

The 10th IEEE Workshop on Silicon Errors in Logic - System Effects, April 2014

Magazine Articles

[PDF]

“Saliency on a chip: a digital approach with an FPGA”

Nachiket Kapre, Dirk Walther, and Christof Koch, and André DeHon

The Neuromorphic Engineer, Volume 1, Issue 2, Autumn 2004

Book Chapters

“Accelerating the SPICE Circuit Simulator using an FPGA - A Case Study”

Nachiket Kapre and André DeHon

From *High-Performance Computing using FPGAs*

Pages 389-427,

Edited by Wim Vanderbauwhede and Khaled Benkrid

Published by *Springer*, Copyright 2013, ISBN-13: 978-1-4614-1790-3

“Programming FPGA Applications in VHDL”

Nachiket Kapre and André DeHon

From *Reconfigurable Computing: The Theory and Practice of FPGA-based Computation*,

Pages 129-153,

Edited by *Scott Hauck* and *André DeHon*,

Published by *Morgan Kaufmann/Elsevier*, Copyright 2008, ISBN-13: 978-0-12-370522-8

Selected Talks

“Machine Learning for FPGA Design” *Invited Talk, ASAP 2025 Special Session on Custom Computing in Canada*, Vancouver, July 2025.

“Machine Learning for FPGA Design” *Invited Talk, Jane Street XCelebrate Colloquium*, New York, May 2025.

“FPGA Architecture and Programming” *Invited Tutorial, SPIN Canada Quantum Computing Workshop*, June 2019.

“A Case for Embedded FPGA-based SoCs for Energy-Efficient Acceleration of Graph Problems”

Pradeep Moorthy, Siddhartha, Nachiket Kapre

Supercomputing Frontiers 2015, March 2015

“SPICE2- A Spatial Parallel Architecture for Accelerating the SPICE Circuit Simulator - Retrospective and Vision” *Talk at Maxeler Inc., University of Glasgow, University of York, Oxford, University of Southampton, National University of Singapore, Mahankorn University of Technology*, 2010-2013.

“Spatial SPICE Mapping and Lessons” Vancouver, Canada *Talk at the University of British Columbia (UBC)*, August 2010.

“Accelerating the SPICE Circuit Simulator using FPGAs” Bengaluru (Bangalore), India *Invited Talk at the Indian Institute of Science (IISC)*, March 2010.

“Accelerating the SPICE Circuit Simulator using FPGAs” Austin, USA *Invited Talk at IBM Inc.*, August 2009.

“Accelerating SPICE Model-Evaluation using FPGAs” San Jose, USA. *Invited Talk at Xilinx Inc.*, February 2009.

“Exploiting Application Structure in On-Chip Network Design” *Invited Talk at University of Gent, Belgium and TU Munich, Germany*, July-August 2007.

Patents

“Coarse grained reconfigurable architecture”,

Nachiket Kapre, Kimon Karras, Dmitri Kitariev, Neil Turton, Siddhartha, Stephan Diestelhorst, and Thilini Dassanayake

U.S. Patent No. 12,585,605 B2, assigned to Xilinx, Inc.; filed Dec 2022, granted Mar 2026.

“Digital circuits for evaluating neural engineering framework style neural networks”,

Benjamin Morcos, Chris Eliasmith, and Nachiket Kapre

U.S. Patent No. 11,537,856 B2, filed Aug 2019, granted Dec 2022.

“Method and a circuit using an associative calculator for calculating a sequence of non-associative operations”,
André DeHon and Nachiket Kapre
U.S. Patent No. 7,991,817 B2, filed Jan 2007, granted Aug 2011.

Advising

Current Waterloo PhD

Emir Guevara (Fall 2025–present)
Ahmed Elsousy (Fall 2025–present; co-supervised with Andrew Boutros)
Thomas Badros (Fall 2025–present; co-supervised with Andrew Boutros)

Current Waterloo MASc

Jehanzeb Mirza (Winter 2023–present; co-supervised with Vincent Gaudet), Queen Elizabeth II Graduate
Scholarship in Science and Technology (QEII-GSST) recipient

Past Waterloo PhD

Gurshaant Malik (2017–2025): *Learning Design Parameters to Build Application Customizable Network-on-Chips for FPGAs*

Past Waterloo MASc

Marat Bekmyrza (2021–2025; co-supervised with Hiren Patel): *Language Model Inference on FPGA with Integer-only Operations*
Aravind Vellora Valayapara (2021–2023; co-supervised with Seyed Zahedi): *Fairness Notions on Hardware Resource Configuration*
MohammadHadi Omid (2021–2024; co-supervised with Seyed Zahedi): *Fair and Efficient Resource Scheduling in Heterogeneous Multi-Agent Systems*
Long Chan Chan (2020–2021): *Implementing FPGA-optimized Systolic Arrays using 2D Knapsack and Evolutionary Algorithms*
Alireza Mellat (2020–2021): *Mocarabe: High-Performance Time-Multiplexed Overlays for FPGAs*
Srinirdheeshwar Kuttuva Prakash (2019–2021): *Managing HBM Bandwidth in Multi-Die FPGAs using Overlay NoCs*
Ian Elmor Lang (2019–2021): *Worst-Case Latency Analysis for the Versal Network-on-Chip*
Benjamin Morcos (2017–2019): *NengoFPGA: an FPGA Backend for the Nengo Neural Simulator*
Tushar Garg (2017–2019): *HopliteBuf FPGA Network-on-Chip: Architecture and Analysis*

NTU PhD

Siddhartha (2013–2019): *Dataflow Optimized Overlays for FPGAs*

NTU MSc

Chethan Kumar Basavaraju (2015): *FPGA NoCs*
Jayakrishnan Selva Kumar (2014): *Maxeler Applications*
Venugopal Swetha (2014): *GPU Monte-Carlo Applications*
Chinnakkannu Adaikkala Raj (2014): *Machine Learning in FPGA CAD*
Jianrong, Kiran Ganapathi, Kunal Gokhale (2014): Miscellaneous topics
Kanchan Kaur, Shipeng Xu (2013): *FPGA Placement/Routing*

NTU UG (Final Year Projects)

Shubham Agarwal (2015): *FPGA NoCs*
Que Yanghua (2015): *Machine Learning*
Dakshina Pradeep Moorthy (2014): *Parallel Graph Accelerators*
Han Jianglei (2014): *Parallel Graph Accelerators*
Soh Jun Jie (2013–14): *Vectorblox*
Favian (2013–14): *3D Convolution using FPGAs*
Lim Hui Hui (2013): *SPICE Fault Tolerance*

Imperial PhD, MSc, MEng, BEng and Interns

Andrew Bean (PhD student 2011–2016): *Adaptive/Learning Systems using FPGAs*
Abid Rafique (PhD student 2010–2013): *Accelerating Semi-Definite Programming with FPGAs, GPUs and*

Multi-Cores

Siddhartha, Dulitha Gunasekara (BEng/MEng students 2011-2012): Various topics

Helene Martorell, Emmanouil Spanakis, Fang Zhou, Wei Lizhong (MSc students 2010-2011): Various topics

Coryan Wilson-Shah (UROP student 2011): Matrix-Free SPICE

Cody Huang (CAPA intern, UC Davis undergraduate 2011): GPU Code-Generation

Caltech Undergraduates and Summer Students

Henry Barnor (2005): *VHDL Design of systolic hardware sorter/placer*

Stephanie Chan (2005, now at NIST): *Experiments on saturating accumulator*

Ravi Teja Sukhavasi (2006, Caltech graduate student): *Applying network-coding ideas to message traffic between parallel compute elements*

Jon Ramirez (2006): *Floating-point associative accumulator*

Corporate Collaborations

Harnhua Ng, Kirvy Teo (Plunify): *Machine-Learning for FPGA CAD*

Jacob Bower (Maxeler): *Maxeler Compiler Framework*

Kumiko Nomura (Toshiba): *Architecture analysis of 3D chips*

Teaching Experience

University of Waterloo

ECE327: *Digital Hardware Systems* (2021, 2023-2025)

ECE493: *Special Topics in Electrical and Computer Engineering* (2025)

ECE627: *Register-transfer-level Digital Systems* (2021, 2023, 2025)

ECE720: *Special Topics in Computers and Digital Systems Software* (2024-2025)

ECE722: *Reconfigurable Computing* (2021, 2023, 2025)

Lecturer

Semester 1 2015, Nanyang Technological University, *CE4052/ES6152: Embedded System Development*

Semester 2 2014, Nanyang Technological University, *CE4054/ES6154: Programmable Systems-on-Chip*

Semester 1 2014, Nanyang Technological University, *CE4052/ES6152: Embedded System Development*

Semester 2 2013, Nanyang Technological University, *CE4054/ES6154: Programmable Systems-on-Chip*

Semester 1 2013, Nanyang Technological University, *CE7451: Research Methods in Computer Science & Engineering*

Semester 1 2013, Nanyang Technological University, *CE4052/ES6152: Embedded Systems Development*

Semester 1 2013, Nanyang Technological University, *ES7501: Electronic Design Automation*

Tutorials/Labs

Semester 1 2015, Nanyang Technological University, *CE3001: Advanced Computer Architecture*

Semester 1 2015, Nanyang Technological University, *CE4052/ES6152: Embedded Systems Development*

Semester 2 2014, Nanyang Technological University, *CE4054/ES6154: Programmable System-on-Chip*

Semester 1 2014, Nanyang Technological University, *CE1005: Digital Logic* (3 groups)

Semester 1 2014, Nanyang Technological University, *CE4052/ES6152: Embedded Systems Development*

Semester 2 2013, Nanyang Technological University, *CE1005: Digital Logic* (1 group)

Semester 1 2013, Nanyang Technological University, *CE4052/ES6152: Embedded Systems Development*

Guest Lectures

Fall 2011, Imperial College London, *ISE2: Computer Architecture*

Winter 2011, Imperial College London, *DoC: Custom Computing*

Teaching Assistant

Spring 2007, University of Pennsylvania, Electrical and Systems Engineering, *ESE680s2: Computer Organization*

Winter 2006, California Institute of Technology, Computer Science, *CS137: Electronic Digital Automation*

Professional Experience

University of Waterloo, Professor (2026-present)

Tenstorrent LLC, Visiting Fellow (2026)

University of Waterloo, Associate Professor (2020-2026)

Xilinx Labs (now AMD Research), Research Director, Singapore (January 2022-February 2023; leave of absence from Waterloo)

University of Waterloo, Assistant Professor (2016-2020)

Nanyang Technological University, Assistant Professor (Oct 2012-Sep 2016)

Plunify, Inc., Chief Technology Officer (2014-2016)

Imperial College London, Junior Research Fellow (October 2010-September 2012)

Maxeler Inc. (now part of Groq, whose inference technology is licensed by NVIDIA), Consultant (July 2011-July 2012)

University of Pennsylvania, Visiting Graduate Student (2006-2010)

Xilinx Inc., Summer Intern (Summer 2005)

Koch Lab (Caltech), Research Assistant (February 2004 to September 2004)

Paxonet Communications Inc. (now Conexant), Employee (August 2002 to August 2003)

Siemens Inc., Part-time Intern (2002).

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